

Rough fractional diffusions as scaling limits of nearly unstable heavy tailed Hawkes processes

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ANALYTICAL ABSTRACT

This paper asks what happens to a Hawkes process, a self-exciting model of order arrivals, when it is nearly unstable (kernel norm just below one) and its kernel has a power-law tail with index $\alpha \in (0, 1)$, so that the mean lag between a trigger and its consequences is infinite. Rescaling space and time with the distance to criticality tied to the horizon, the authors show that the cumulated count converges to a random, non-semimartingale limit. For $\alpha > 1/2$ this limit is differentiable, and its derivative solves a fractional Cox–Ingersoll–Ross equation driven by a Mittag-Leffler kernel, with Hölder regularity $\alpha - 1/2$. A light tail, by contrast, gives the classical Brownian CIR limit. The financial reading is that persistence in signed order flow, created by order splitting, together with the high endogeneity of modern markets, can generate the very rough volatility measured empirically, an agent-based foundation for rough volatility models. The result is a theorem, not an estimate: it contains no data, and the link to market volatility is an interpretation. This review reconstructs the proofs, checks the limit kernel and sample-path randomness by simulation, and assesses the coupled asymptotic regime as the main weakness.

1 Introduction

1.1 What the paper is about

PAPER Jaisson and Rosenbaum (2016) study a sequence of Hawkes processes, that is, self-exciting point processes in which every event raises the arrival rate of future events, whose kernel has an L^1 norm just below one (the process is “nearly unstable”) and a power-law tail $x^{-(1+\alpha)}$ with $\alpha \in (0, 1)$. They prove that, after a suitable rescaling of space and time, the process converges in law to a limit that is neither a diffusion nor a semimartingale. When $\alpha \in (1/2, 1)$ the limit is the time integral of a fractional Cox–Ingersoll–Ross type process whose paths have Hölder regularity $\alpha-1/2$, the roughness that Gatheral, Jaisson and Rosenbaum (2018) measured in realised volatility.

CRITIQUE The paper is a result in probability, not in empirical finance. Its financial content is an interpretation: the Hawkes count stands for order flow, the limit of the derivative stands for volatility, and the exponent α stands for the persistence of signed order flow created by order splitting. The review therefore treats the theorem as the evidence and the financial reading as an inference that the theorem makes available but does not itself establish.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

Whether the very irregular volatility found in high-frequency data can be generated by a microscopic, agent-based mechanism rather than postulated as a fractional Brownian input, and, mathematically, what the scaling limit of a heavy-tailed nearly unstable Hawkes process is.

WHY DOES THE QUESTION MATTER?

Before 2014 the standard stochastic volatility models were Brownian diffusions, with a Hurst-type regularity of one half. A volatility path with Hölder exponent near 0.1 changes option pricing, hedging, and the forecasting of realised variance. A rough volatility model fitted without a microscopic justification is a statistical description; a derivation from order flow gives it a reason to exist and ties its parameter to an observable behaviour.

WHAT WAS KNOWN BEFORE?

Light-tailed nearly unstable Hawkes processes converge to an integrated Brownian CIR process (Jaisson and Rosenbaum 2015). Empirical calibrations of Hawkes processes to order flow return kernels with norm close to one (Filimonov and Sornette 2012; Hardiman, Bercot and Bouchaud 2013) and a slowly decaying kernel tail (Bacry, Jaisson and Muzy 2014, cited by the authors). Signed order flow has long memory because of metaorder splitting (Lillo and Farmer 2004).

WHAT IS MISSING FROM THE LITERATURE?

The scaling limit when $\int_0^\infty s\phi(s) ds = \infty$, where the semimartingale machinery of the earlier paper does not apply, and an agent-based account of rough volatility.

WHAT DOES THE PAPER PROPOSE, AND HOW DOES IT INVESTIGATE IT?

A coupled scaling regime in which the observation horizon T , the distance to criticality $1 - a_T$ and the baseline intensity μ_T move together, followed by a proof of tightness, an identification of limit points through a Volterra equation, and a fractional-calculus argument that yields the derivative and its regularity.

WHAT DOES IT FIND, AND WHY DOES IT MATTER?

Limit points (Z, X) satisfy $Z = B_X$ and a Volterra equation with the Mittag-Leffler kernel $f^{\alpha, \lambda}$; for $\alpha > 1/2$ the limit X is differentiable and its derivative solves a fractional CIR equation with Hölder exponent $\alpha - 1/2$. The finding matters because fat tails, which suggest persistence, produce an irregular limit; aggregation of persistent shocks yields roughness.

2 Research Problem and Hypothesis

2.1 Research question

PAPER Let N^T be a Hawkes process with intensity $\lambda_t^T = \mu^T + \int_0^t \phi^T(t-s) dN_s^T$, $\phi^T = a_T \phi$, $\|\phi\|_1 = 1$, $a_T \uparrow 1$, and $\alpha x^\alpha(1-F(x)) \rightarrow K$ as $x \rightarrow \infty$. The question is: what is the law of the weak limit of $X_t^T = \frac{1-a_T}{T^\alpha \mu^\alpha \delta^{-1}} N_{Tt}^T$ on $[0, 1]$ as $T \rightarrow \infty$, and how smooth are its sample paths and its derivative?

2.2 Motivation

PAPER Two empirical regularities motivate the question: the fitted L^1 norms of Hawkes kernels on trade and quote data are close to one, which the authors read as a high degree of endogeneity in modern markets, and the fitted kernels have power-law tails, which they relate to order splitting. A third regularity, the linear relation between cumulated order flow and integrated variance (Wyart et al. 2008, cited by the authors), makes the derivative of the Hawkes count a natural candidate for variance.

2.3 Existing literature and research gap

PAPER The gap is the heavy-tailed case. In Jaisson and Rosenbaum (2015) the condition $\int s\phi(s)ds < \infty$ drives a CIR limit through semimartingale convergence theorems. Without it the sequence of

rescaled intensities is not tight, so the intensity route is closed and the authors work with the counting process directly, in the spirit of Zhu (2013).

2.4 Hypothesis and contribution

INFERENCE The paper states no statistical hypothesis. Its implicit hypothesis is a conditional statement: if order flow is a Hawkes process with norm near one and tail index α , then variance is rough with $H = \alpha - 1/2$. The contribution claimed is the first agent-based foundation for rough volatility; what is demonstrated is the limit theorem. The step from the theorem to market microstructure is plausible and consistent with the evidence the authors cite, but it is not tested in the paper (Section 7).

3 Theoretical Framework and Mechanism

3.1 The Hawkes model and the stability condition

PAPER A Hawkes process has intensity

$$\lambda_t = \mu + \sum_{0 < J_i < t} \phi(t - J_i) = \mu + \int_{(0,t)} \phi(t - s) dN_s, \quad (1)$$

N_t Counting process of events (orders, trades) up to time t .

λ_t Conditional arrival rate of new events given the past.

$\mu > 0$ Exogenous (immigrant) arrival rate.

$\phi \geq 0$ Kernel: extra rate generated at lag $t - J_i$ by an event at J_i .

J_i Event times before t .

Each event triggers new events at rate ϕ , so on average it has $\|\phi\|_1 = \int_0^\infty \phi(s) ds$ direct offspring. The process is a branching process with immigration. If $\|\phi\|_1 < 1$ the branching is subcritical and, in the stationary regime, $\mathbb{E}[\lambda_t] = \mu/(1 - \|\phi\|_1)$: the share of events that are endogenous (triggered by other events) is $\|\phi\|_1$. This is the sense in which a norm near one signals a market in which most orders are reactions to other orders. The condition $\|\phi\|_1 < 1$ plays the role of $|\rho| < 1$ in an AR(1) model, and for the same reason the process then looks like a Poisson process at large scale.

3.2 Nearly unstable heavy-tailed sequences

PAPER The paper works with a triangular array. For each horizon T the intensity is $\lambda_t^T = \mu^T + \int_0^t \phi^T(t - s) dN_s^T$ with $\phi^T = a_T \phi$.

Table 1. Assumptions of the main theorems

Assumption	Formal statement	Meaning
Near criticality	$a_T < 1, a_T \rightarrow 1, \ \phi\ _1 = 1$	Subcritical for each T ; critical in the limit
Heavy tail (A1)	$\alpha x^\alpha(1 - F(x)) \rightarrow K, \alpha \in (0, 1)$	Kernel tail $\sim Kx^{-(1+\alpha)}$; infinite mean lag
Critical rate (A2)	$T^\alpha(1 - a_T) \rightarrow \lambda\delta$	Distance to criticality shrinks like $T^{-\alpha}$
Baseline (A2)	$T^{1-\alpha}\mu_T \rightarrow \mu^*\delta^{-1}$	Immigration rate shrinks like $T^{\alpha-1}$
Constant	$\delta = K\Gamma(1 - \alpha)/\alpha$	Scale of the Laplace expansion of ϕ

Source: Jaisson and Rosenbaum (2016), Assumptions 1 and 2, reorganised by the reviewer.

The two scalings in Assumption 2 are not arbitrary. The authors derive them from the requirement that the limit be random but approximately stationary, as follows.

3.3 The intensity as a Volterra equation

PAPER Let $M_t^T = N_t^T - \int_0^t \lambda_s^T ds$ be the compensated martingale and let

$$\psi^T(t) = \sum_{k=1}^{\infty} (\phi^T)^{*k}(t), \quad (2)$$

$(\phi^T)^{*k}$ The k -fold convolution of ϕ^T with itself.

ψ^T Resolvent kernel: expected number of descendants, over all generations, at lag t from a given event.

The series converges because $\|\phi^T\|_1 < 1$. Inverting the Volterra equation for the intensity gives

$$\lambda_{tT}^T = \mu^T + \int_0^{tT} \psi^T(Tt - s) \mu^T ds + \int_0^{tT} \psi^T(Tt - s) dM_s^T. \quad (3)$$

The first integral is the mean response to the immigrants; the second is the response to martingale surprises. With the space scaling $(1 - a_T)/\mu^T$, which is natural because $\mathbb{E}[\lambda_t] = \mu^T/(1 - a_T)$ in the stationary case, the rescaled intensity $C_t^T = \frac{1-a_T}{\mu^T} \lambda_{tT}^T$ satisfies

$$C_t^T = (1 - a_T) + \int_0^t T(1 - a_T) \psi^T(Ts) ds + \sqrt{\frac{T(1-a_T)}{\mu^T}} \int_0^t \psi^T(T(t-s)) \sqrt{C_s^T} dB_s^T, \quad (4)$$

with $B_t^T = T^{-1/2} \int_0^{tT} dM_s^T / \sqrt{\lambda_s^T}$. Equation (4) says that the intensity is a deterministic drift plus a stochastic convolution of the square root of itself against ψ^T : a Volterra analogue of the CIR equation $dC = \kappa(\theta - C)dt + \sigma\sqrt{C}dB$. Its limit is therefore governed by the limit of $x \mapsto \psi^T(Tx)$.

3.4 The key computation: a geometric random sum and the Mittag-Leffler function

PAPER Normalising ψ^T gives a probability density,

$$\rho^T(x) = T \frac{\psi^T(Tx)}{\|\psi^T\|_1}, \quad J^T = \frac{1}{T} \sum_{i=1}^{I^T} X_i, \quad \mathbb{P}[I^T = k] = (1 - a_T) a_T^{k-1}, \quad (5)$$

where ρ^T is the density of J^T , the X_i are i.i.d. with density ϕ , and I^T is geometric. The Laplace transform of J^T , writing $\hat{\phi}$ for that of ϕ , is

$$\hat{\rho}^T(z) = \sum_{k \geq 1} (1 - a_T) a_T^{k-1} \hat{\phi}\left(\frac{z}{T}\right)^k = \frac{\hat{\phi}(z/T)}{1 - \frac{a_T}{1-a_T} (\hat{\phi}(z/T) - 1)}. \quad (6)$$

Integration by parts gives $\hat{\phi}(z) = 1 - z \int_0^\infty e^{-zt} (1 - F(t)) dt$, and Karamata's Tauberian theorem applied to the regularly varying tail yields

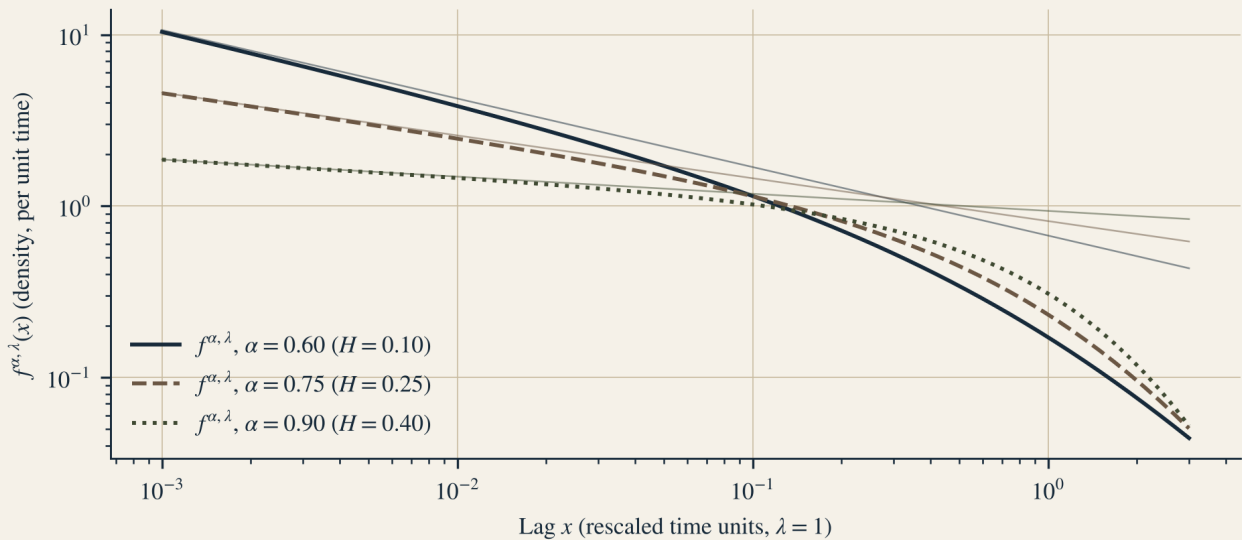
$$\hat{\phi}(z) = 1 - \delta z^\alpha + o(z^\alpha), \quad \delta = K \frac{\Gamma(1 - \alpha)}{\alpha}. \quad (7)$$

Substituting $z \rightarrow z/T$ in (6), the denominator becomes $1 + \frac{a_T}{1-a_T} \delta T^{-\alpha} z^\alpha \rightarrow 1 + z^\alpha/v_T$ with $v_T = \delta^{-1} T^\alpha (1 - a_T)$, so that

$$\hat{\rho}^T(z) \sim \frac{v_T}{v_T + z^\alpha}. \quad (8)$$

This is the Laplace transform of $v x^{\alpha-1} E_{\alpha,\alpha}(-v x^\alpha)$, where $E_{\alpha,\beta}(z) = \sum_{n \geq 0} z^n / \Gamma(\alpha n + \beta)$ is the Mittag-Leffler function. The reviewer checks this inversion with the standard identity $\mathcal{L}\{x^{\beta-1} E_{\alpha,\beta}(-c x^\alpha)\}(z) = z^{\alpha-\beta} / (z^\alpha + c)$ at $\beta = \alpha$. The economically important point is the behaviour of the limit kernel near zero, $f^{\alpha,\lambda}(x) \sim \lambda x^{\alpha-1} / \Gamma(\alpha)$ (Proposition 3.1): the singularity at the origin is created by the aggregation of an infinite sum of heavy-tailed lags, not by any singularity of ϕ itself.

Figure 1. The limit kernel $f^{\alpha,\lambda}(x) = \lambda x^{\alpha-1} E_{\alpha,\alpha}(-\lambda x^\alpha)$ for three tail indices



Source: reviewer's computation (generated, not from the paper), $\lambda = 1$, Mittag-Leffler function evaluated by series at 40-digit precision; $x \in [10^{-3}, 3]$. Thin lines are the small- x asymptote $x^{\alpha-1} / \Gamma(\alpha)$ of Proposition 3.1. Both axes logarithmic.

Figure 1 shows the singular behaviour at the origin. On log-log axes the kernel is a straight line of slope $\alpha - 1$ for small x and bends downward once λx^α is of order one. The smaller α , the steeper the singularity, the closer the kernel is to non-integrable, and the rougher the process it generates.

3.5 Choosing the regime: why Assumption 2

PAPER Inserting the Mittag-Leffler kernel in (4) gives, heuristically for $\alpha > 1/2$,

$$C_t^\infty \sim v_\infty \int_0^t s^{\alpha-1} E_{\alpha,\alpha}(-v_\infty s^\alpha) ds + \gamma_\infty v_\infty \int_0^t (t-s)^{\alpha-1} E_{\alpha,\alpha}(-v_\infty (t-s)^\alpha) \sqrt{C_s^\infty} dB_s, \quad (9)$$

with $\gamma_T = 1/\sqrt{\mu^T T(1-a_T)}$. A random limit requires v_∞ and γ_∞ both to be finite and positive (or $v_\infty = 0$ with $\gamma_\infty v_\infty > 0$, which makes $\mathbb{E}[(C_t^\infty)^2]$ grow like $t^{2\alpha-1}$ and is rejected as incompatible with approximate stationarity). Hence Assumption 2. The reviewer verifies the constants: with $\mu_T \sim \mu^* \delta^{-1} T^{\alpha-1}$ and $T(1-a_T) \sim \lambda \delta T^{1-\alpha}$, the product $\mu^T T(1-a_T) \rightarrow \mu^* \lambda$, so $\gamma_\infty = 1/\sqrt{\mu^* \lambda}$, which is exactly the constant in Theorems 3.1 and 3.2. **CRITIQUE** The argument for rejecting the other regimes is a plausibility argument on second moments, not a theorem; the regime is chosen, not derived.

3.6 Main results

PAPER Define the rescaled count, integrated intensity and martingale on $[0, 1]$:

$$X_t^T = \frac{1-a_T}{T^\alpha \mu^* \delta^{-1}} N_{Tt}^T, \quad \Lambda_t^T = \frac{1-a_T}{T^\alpha \mu^* \delta^{-1}} \int_0^{tT} \lambda_s^T ds, \quad Z_t^T = \sqrt{\frac{T^\alpha \mu^* \delta^{-1}}{1-a_T}} (X_t^T - \Lambda_t^T). \quad (10)$$

PROPOSITION 3.2

Under Assumptions 1 and 2 the sequence (Z^T, X^T) is tight. Every limit point (Z, X) has Z a continuous martingale with $[Z, Z] = X$.

THEOREM 3.1

There is a Brownian motion B such that $Z_t = B_{X_t}$, X is continuous with Hölder regularity $(1 \wedge 2\alpha) - \varepsilon$ for every $\varepsilon > 0$, and

$$X_t = \int_0^t s f^{\alpha,\lambda}(t-s) ds + \frac{1}{\sqrt{\mu^* \lambda}} \int_0^t f^{\alpha,\lambda}(t-s) B_{X_s} ds. \quad (11)$$

THEOREM 3.2

Let $\alpha > 1/2$ and X satisfy (11). Then X is differentiable, with derivative Y solving

$$Y_t = F^{\alpha,\lambda}(t) + \frac{1}{\sqrt{\mu^*\lambda}} \int_0^t f^{\alpha,\lambda}(t-s) \sqrt{Y_s} dB_s^1, \quad (12)$$

where $F^{\alpha,\lambda}(t) = \int_0^t f^{\alpha,\lambda}(s) ds = 1 - E_{\alpha,1}(-\lambda t^\alpha)$ and B^1 is a Brownian motion. For every $\varepsilon > 0$, Y has Hölder regularity $\alpha - 1/2 - \varepsilon$.

Equation (11) says that the limit count equals a deterministic ramp, the cumulated mean response $\int_0^t F^{\alpha,\lambda}(s) ds$, plus a convolution of the Mittag-Leffler kernel with the time-changed Brownian motion B_X . The clock of the Brownian motion is X itself: when activity is high, noise accumulates faster, which is the self-referential character that makes the equation a Volterra analogue of the CIR square-root diffusion. Equation (12) is the same statement in terms of the activity rate $Y = dX/dt$. Because $f^{\alpha,\lambda}(x) \sim x^{\alpha-1}$ is the Volterra kernel of a fractional Brownian motion with $H = \alpha - 1/2$ (the Mandelbrot and Van Ness (1968) representation, whose kernel is $(t-s)^{H-1/2}$), the authors read (12) as a fractional CIR process with that Hurst parameter. The mean of Y_t is $F^{\alpha,\lambda}(t)$, which rises from zero: the limit starts from an empty past and is not stationary.

3.7 Reconstruction of the proofs

PAPER The reviewer reconstructs the main steps, as the proofs carry the mechanism.

STEP 1: TIGHTNESS (PROPOSITION 3.2)

From Bacry et al. (2013), $\mathbb{E}[N_t^T] = \mu_T t + \mu_T \int_0^t \psi^T(t-s) s ds \leq t\mu_T(1 + \|\psi^T\|_1)$, and $\|\psi^T\|_1 \leq c/(1-a_T)$, so $\mathbb{E}[X_1^T] = \mathbb{E}[\Lambda_1^T] \leq c$. Both processes are increasing and their maximal jump $(1-a_T)/T^\alpha \rightarrow 0$, so they are $\mathbb{C}$ -tight (Jacod and Shiryaev 2003, Prop. VI-3.26). The predictable bracket of Z^T is Λ^T , hence Z^T is tight, any limit Z is continuous, and $[Z, Z] = X$ because $[Z^T, Z^T] = X^T$. Doob's inequality gives

$$\mathbb{E} \left[\sup_{t \leq 1} (X_t^T - \Lambda_t^T)^2 \right] \leq c \left(\frac{1-a_T}{T^\alpha} \right)^2 \mathbb{E}[N_T^T] \leq c \frac{1-a_T}{T^\alpha} \rightarrow 0, \quad (13)$$

so working with Λ^T instead of X^T costs nothing in the limit.

STEP 2: THE VOLTERRA IDENTITY (EQUATION (11))

Using $\psi^T * \phi^T = \psi^T - \phi^T$ the cumulated intensity satisfies

$$\int_0^t \lambda_s^T ds = t\mu_T + \int_0^t \psi^T(t-s) s \mu_T ds + \int_0^t \psi^T(t-s) M_s^T ds. \quad (14)$$

After the change $t \rightarrow Tt$ and the normalisation, $\Lambda_t^T = T_1 + T_2 + T_3$ with $T_1 = (1-a_T)t\mu_T \rightarrow 0$ (where $\mu_T = \mu_T/(\mu^* \delta^{-1} T^{\alpha-1}) \rightarrow 1$), $T_2 = a_T \mu_T \int_0^t F^T(t-s) ds \rightarrow \int_0^t F^{\alpha,\lambda}(t-s) ds = \int_0^t f^{\alpha,\lambda}(t-s) ds$ by the weak convergence

of ρ^T (Lemma 4.3), and

$$T_3 = \frac{a_T}{\sqrt{T^\alpha(1-a_T)\mu^*\delta^{-1}}} \int_0^t \rho^T(t-s)Z_s^T ds \rightarrow \frac{1}{\sqrt{\mu^*\lambda}} \int_0^t f^{\alpha,\lambda}(t-s)Z_s ds. \quad (15)$$

The convergence of T_3 rests on the pathwise integration by parts $\int_0^t \rho^T(t-s)Z_s^T ds = \int_0^t F^T(t-s)dZ_s^T$, the Itô isometry bound $\mathbb{E}[(\int_0^t (F^{\alpha,\lambda} - F^T)(t-s)dZ_s^T)^2] \leq c \int_0^t (F^{\alpha,\lambda} - F^T)^2(t-s)ds \rightarrow 0$, and almost sure convergence on a Skorokhod representation space. The identity $Z = B_X$ is Dambis–Dubins–Schwarz.

STEP 3: HÖLDER REGULARITY BY BOOTSTRAP (LEMMA 4.4)

If X is H -Hölder then $Z = B_X$ is $(H/2 - \varepsilon)$ -Hölder (Brownian regularity composed with a time change of regularity H). The convolution of $f^{\alpha,\lambda}$, whose fractional integral $I^\nu f^{\alpha,\lambda}$ has a singularity $x^{\alpha+\nu-1}$, with the fractional derivative of Z then improves the regularity of X to $(\alpha + H/2) \wedge 1$. This defines a map $H \mapsto \alpha + H/2$ whose fixed point is $H^* = 2\alpha$ and which contracts with factor $1/2$:

$$H_0 = \alpha, \quad H_{n+1} = \min(1, \alpha + \frac{H_n}{2}), \quad H_n \rightarrow \min(1, 2\alpha). \quad (16)$$

Starting from $H_0 = \alpha$ (Proposition A.3 with Proposition 3.1) and arguing by contradiction on the supremum of attainable exponents, the authors obtain $(1 \wedge 2\alpha) - \varepsilon$. This is the exponent of Theorem 3.1.

STEP 4: DIFFERENTIATION BY FRACTIONAL CALCULUS (THEOREM 3.2)

For $\nu \in (0, \alpha)$, the fractional integration by parts of Corollary A.2, together with Proposition 3.1, allows Equation (11) to be rewritten as

$$X_t = \int_0^t s f^{\alpha,\lambda}(t-s) ds + \frac{1}{\sqrt{\mu^*\lambda}} \int_0^t D^\nu f^{\alpha,\lambda}(t-s) I^\nu Z_s ds, \quad (17)$$

and for $\nu > 1/2$, $I^\nu Z_s = \int_0^s D^{1-\nu} Z_u du$, so Fubini gives $X_t = \int_0^t Y_s ds$ with

$$Y_s = F^{\alpha,\lambda}(s) + \frac{1}{\sqrt{\mu^*\lambda}} \int_0^s D^\nu f^{\alpha,\lambda}(s-u) D^{1-\nu} Z_u du. \quad (18)$$

Since $D^\nu f^{\alpha,\lambda}(x) \sim x^{-(1-\alpha+\nu)}$, Proposition A.3 with $\beta = 1 - \alpha + \nu$ gives Hölder regularity $\alpha - \nu$ for Y , and letting $\nu \downarrow 1/2$ gives $\alpha - 1/2 - \varepsilon$. The stochastic Fubini theorem (Veraar 2012) converts $D^{1-\nu} Z$ into a stochastic integral, and the identity $I^{\nu+1} D^\nu f = I^\nu f$ collapses the expression to $\int_0^t f^{\alpha,\lambda}(t-u) dZ_u$. A final representation theorem (Revuz and Yor 1999, Thm. V-3.8) writes $Z = \int \sqrt{Y} dB^1$, which is (12). The interval for ν is non-empty exactly when $1/2 < \nu < \alpha$, that is, when $\alpha > 1/2$. This is the source of the threshold: $D^{1-\nu} Z$ exists only if $1 - \nu < 1/2$ (the regularity of the martingale Z) and $D^\nu f^{\alpha,\lambda}$ exists only if $\nu < \alpha$.

3.8 Economic reading and what is testable

INFERENCE Three predictions follow if the Hawkes representation of order flow is accepted. First, the roughness of variance and the tail index of the kernel are linked by $H = \alpha - 1/2$, so an estimate of either identifies the other. The authors state this in their discussion, noting that α is hard to measure while smoothness is easy. Second, roughness is a near-critical phenomenon: the relevant observation window scales like $(1 - \|\phi\|_1)^{-1/\alpha}$, so a market far from criticality shows classical behaviour at ordinary horizons. Third, the volatility must have a deterministic mean that increases from the origin, an artefact of starting from an empty past.

Table 2. Observation window at which roughness appears, $(1 - \|\phi\|_1)^{-1/\alpha}$, in units of the kernel's time scale

Tail index α	$H = \alpha - 1/2$	$\Gamma(1 - \alpha)$	$\ \phi\ _1 = 0.97$	$\ \phi\ _1 = 0.99$
0.55	0.05	1.97	587	4,329
0.60	0.10	2.22	345	2,154
0.75	0.25	3.63	107	464
0.90	0.40	9.51	49	167

Source: reviewer's calculation from the scaling stated in the paper's discussion; an illustration, not a result of the paper. The unit is the time scale on which the kernel is defined (for example, the lower cutoff of a Pareto kernel). Values rounded.

Table 2 shows why the reading is delicate. For an empirical $H \approx 0.1$ (the value reported by Gatheral et al. 2018), the implied tail index is $\alpha \approx 0.6$, and a kernel norm of 0.99 makes the window thousands of kernel time units long. Whether intraday data at realistic horizons fall in the asymptotic window is not addressed by the paper.

4 Data and Empirical Strategy

4.1 What the paper uses

PAPER The paper contains no data set, no estimator, no regression and no table or figure. Its empirical content is entirely imported: the near-unit kernel norms and power-law kernel tails reported for order-flow Hawkes fits (Filimonov and Sornette 2012; Hardiman, Bercot and Bouchaud 2013; Bacry, Jaisson and Muzy 2014), the linear relation between cumulated order flow and integrated variance (Wyart et al. 2008), and the smoothness estimates of Gatheral, Jaisson and Rosenbaum (2018). The sections of the review on data, variable construction, identification and sample bias therefore apply to the cited evidence, not to this paper, and are not reconstructed here.

EVIDENCE The identifying variation of a theorem is its hypotheses. The only place where the result could be wrong is in a gap between what the hypotheses assume and what is true of markets. This is addressed in the critique.

4.2 Numerical checks generated for this review

EXTENSION To test the mechanics of the theorem, the reviewer simulated the model. The Hawkes process has the Pareto kernel $\phi(x) = \alpha x^{-(1+\alpha)} \mathbf{1}_{x \geq 1}$, so that $1 - F(x) = x^{-\alpha}$, $K = \alpha$ and $\delta = \Gamma(1 - \alpha)$. Parameters: $\alpha = 0.75$, $\lambda = 1$, $\mu^* = 1$. Two experiments were run.

Table 3. Design of the numerical checks (generated by the reviewer)

Item	Experiment A: law of J^T	Experiment B: sample paths
Question	Does ρ^T converge to $f^{\alpha, \lambda}$?	Is the limit random, and is the mean ramp visible?
Horizon T	10^4	2×10^4
Branching parameter a_T	$1 - \lambda \delta T^{-\alpha} = 0.99637$	$1 - \lambda \delta T^{-\alpha} = 0.99784$
Baseline μ_T	not needed	$\mu^* \delta^{-1} T^{\alpha-1} = 0.0232$
Method	60,000 draws of $J^T = T^{-1} \sum_{i \leq T} X_i$	Cluster (branching) simulation, 5 seeds

Source: reviewer's design. Cluster simulation: immigrants arrive as a Poisson process of rate μ_T on $[0, T]$, and each event has a $\text{Poisson}(a_T)$ number of children displaced by independent Pareto lags.

The cluster representation is exact for the Hawkes process (Hawkes and Oakes 1974), so Experiment B simulates the process itself rather than an approximation to it. Experiment A samples the random variable whose law is the object of Section 3.4.

5 Results

5.1 Result 1: the limit kernel and the regime

EVIDENCE The law of J^T converges to the density $f^{\alpha, \lambda}(x) = \lambda x^{\alpha-1} E_{\alpha, \alpha}(-\lambda x^\alpha)$ (Lemma 4.3), with $f^{\alpha, \lambda}(x) \sim \lambda x^{\alpha-1} / \Gamma(\alpha)$ at zero. The mass of this density on $[0, 1]$ is $F^{\alpha, \lambda}(1) = 1 - E_{\alpha, 1}(-\lambda)$, which equals 0.608 for $\alpha = 0.75$, $\lambda = 1$ (reviewer's computation). The interpretation is that after an event, a fraction of about 61% of its total (all-generation) response is delivered within a time T , the horizon, while the rest arrives later: the response is spread over scales much longer than the observation window, which is what the heavy tail does.

5.2 Result 2: the limit is random and non-semimartingale

EVIDENCE Theorem 3.1 gives the limit X Hölder regularity $(1 \wedge 2\alpha) - \varepsilon$. For $\alpha > 1/2$ this is almost one, so X itself is smooth; its derivative Y is not. The authors stress that, unlike earlier Hawkes scaling limits, the heavy-tailed limit is not an integrated semimartingale, which is why semimartingale convergence theorems cannot be used and the proof proceeds through fractional calculus.

5.3 Result 3: the roughness of the activity rate

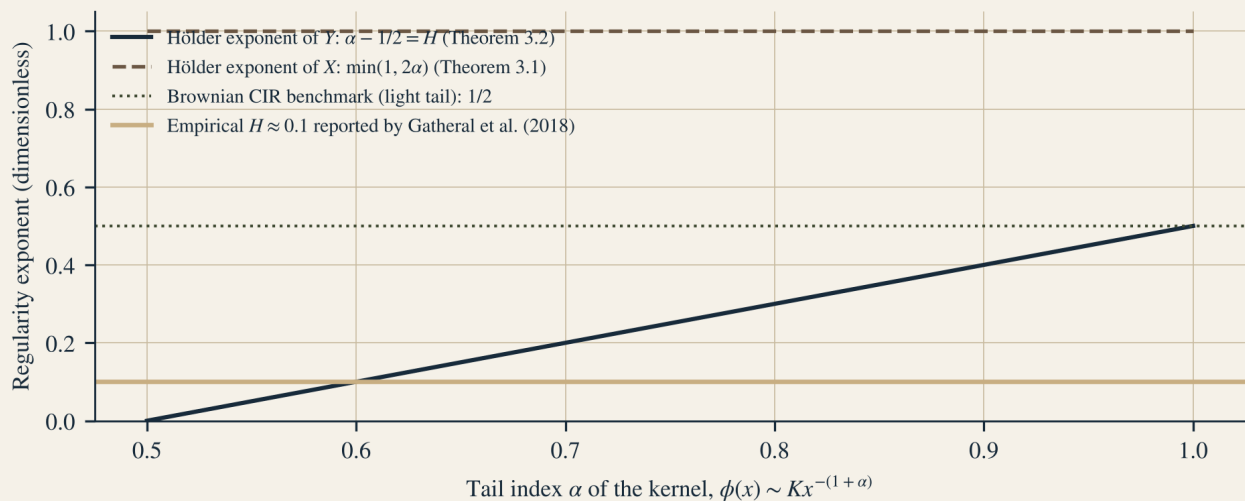
EVIDENCE The exponent of Y is $H = \alpha - 1/2$. The table below translates it.

Table 4. Regularity exponents implied by the theorems

Tail index α	$X: 1 \wedge 2\alpha$	$Y: \alpha - 1/2$	Comment
$\alpha \leq 1/2$	2α	not defined	X not differentiable; Theorem 3.2 does not apply
0.55	1	0.05	Extremely rough
0.60	1	0.10	Matches the empirical value reported by Gatheral et al. (2018)
0.75	1	0.25	Rough, smoother than empirical
0.90	1	0.40	Close to Brownian (0.5)
Light tail ($\int s\phi < \infty$)	1	$1/2$	Brownian CIR (Jaisson and Rosenbaum 2015)

Source: Theorems 3.1 and 3.2 of the paper; the $\alpha \leq 1/2$ row follows from the statement of Theorem 3.1, and the last row from the paper's Introduction. The Hölder exponents are almost sure, up to an arbitrary $\varepsilon > 0$.

Figure 2. Hölder exponents of the limit count X and of its derivative Y as functions of the kernel tail index α



Source: Theorems 3.1 and 3.2 of Jaisson and Rosenbaum (2016), plotted by the reviewer (generated); empirical reference from Gatheral et al. (2018) as cited in the paper. Exponents hold up to an arbitrarily small ε ; $\alpha \in [0.5, 1]$.

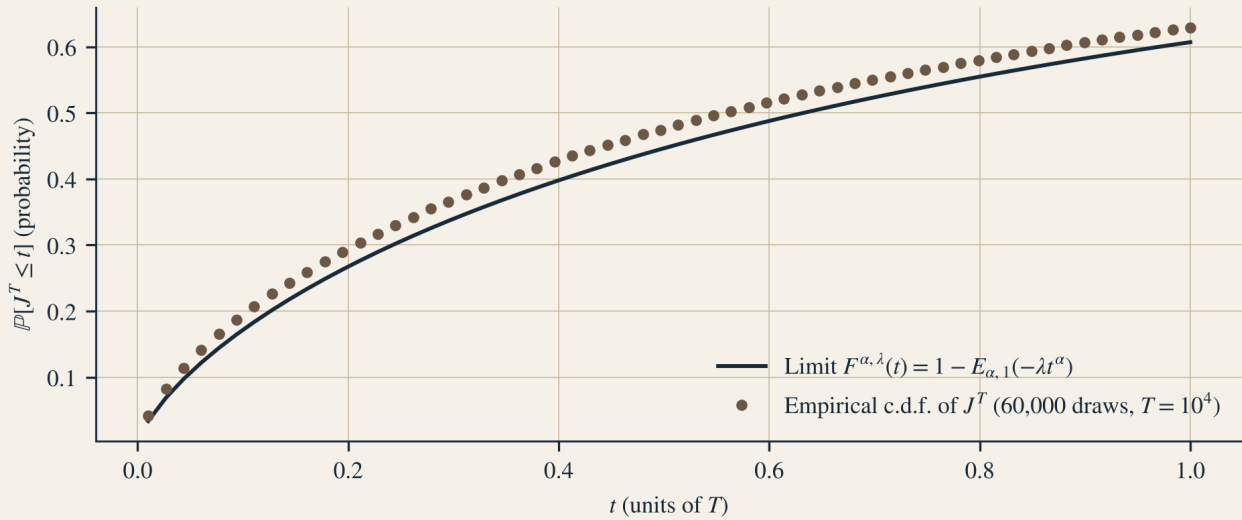
Figure 2 shows the central qualitative claim. The regularity of Y falls linearly to zero as $\alpha \downarrow 1/2$, while that of X stays at one: the cumulated count is smooth, the rate at which it accumulates is rough. For $\alpha \rightarrow 1$ the exponent approaches $1/2$ and the classical Brownian CIR behaviour is recovered, which is a consistency check with the light-tailed result, although at $\alpha = 1$ the tail assumption $\alpha \in (0, 1)$ is violated and the theorem itself says nothing.

5.4 Numerical evidence

EVIDENCE Figure 3 compares the empirical distribution of J^T with the limit $F^{\alpha, \lambda}$. The maximum absolute difference over $t \in [0.01, 1]$ is 0.031, the empirical distribution being everywhere above the limit by one to three percentage points; with 60,000 draws the Monte Carlo standard error is below 0.002, so the

gap is a finite- T effect rather than noise. The reviewer did not isolate its source. Plausible candidates are the $o(z^\alpha)$ remainder in the Laplace expansion, which for a Pareto kernel with lower cutoff one is of order z , and the finite value $a_T = 0.9964$. The sign and size of the gap are consistent with convergence but a larger T would be needed to confirm the rate.

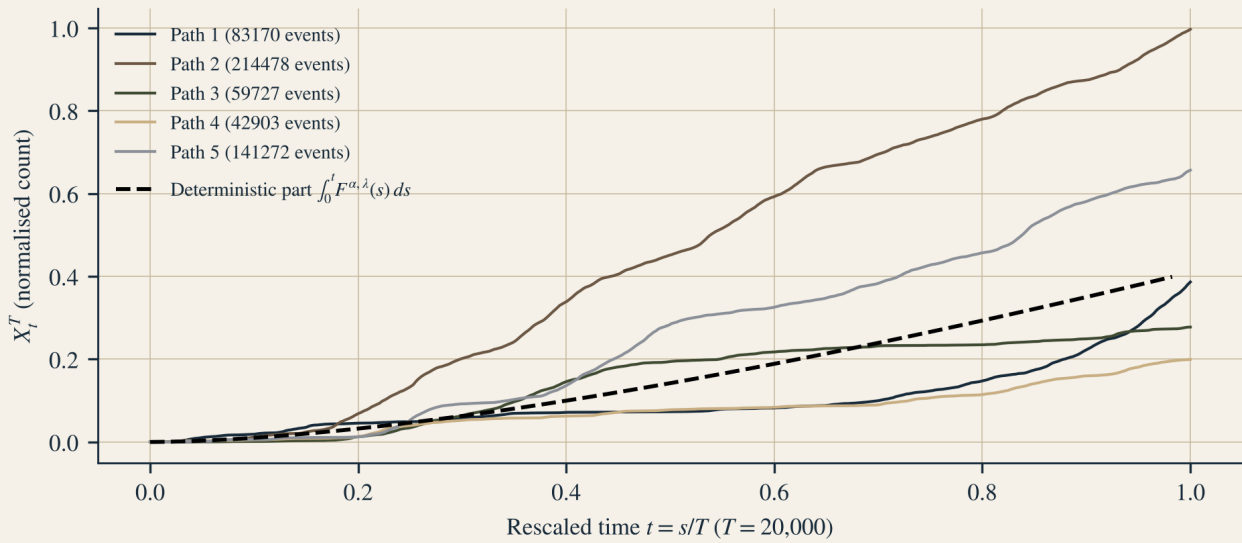
Figure 3. Empirical distribution of the rescaled family time J^T against the limit $F^{\alpha,\lambda}(t) = 1 - E_{\alpha,1}(-\lambda t^\alpha)$



Source: reviewer's Monte Carlo (generated, not from the paper). $\alpha = 0.75$, $\lambda = 1$, $T = 10^4$, $a_T = 0.99637$, 60,000 draws of J^T with Pareto lags of lower cutoff one; limit evaluated by series.

Figure 4 shows five independent simulated paths of the rescaled count X^T . The total number of events over $[0, T]$ varies from 42,903 to 214,478, a five-fold spread across seeds with identical parameters, and the paths cross the deterministic ramp $\int_0^t F^{\alpha,\lambda}(s)ds$ in both directions. This is the empirical signature of the claim that the limit is not degenerate: near criticality, endogenous clustering makes the cumulated activity random at first order, so that the law of large numbers fails.

Figure 4. Rescaled cumulated event count X_t^T for five simulated nearly unstable heavy-tailed Hawkes processes, and the deterministic mean ramp of Theorem 3.1



Source: reviewer's simulation (generated, not from the paper). $\alpha = 0.75$, $\lambda = \mu^* = 1$, $T = 20,000$, $1 - a_T = 0.00216$, $\mu_T = 0.0232$, Pareto kernel with cutoff one; five seeds. The dashed line is the first term of Equation (11).

5.5 Economic significance

INFERENCE The result is economically relevant only through the interpretation. If $\alpha \approx 0.6$ and $\|\phi\|_1$ is within a few percent of one, then volatility measured over a window of order $(1 - \|\phi\|_1)^{-1/\alpha}$ kernel units is rough with $H \approx 0.1$. The theorem converts a statement about order-flow persistence into one about option-relevant quantities: the roughness of variance determines the power-law behaviour of the at-the-money implied volatility skew in models of the rough Heston type (El Euch and Rosenbaum 2019, in the library of this project). No coefficient, standard error or t -statistic exists in the paper, and none is claimed here.

6 Robustness and Additional Tests

CRITIQUE A theoretical paper has no robustness section in the econometric sense. The relevant questions are whether the conclusion survives relaxing the assumptions and whether the proof has hidden steps. The authors discuss four relaxations in their Section 3.4. The table follows the format concern, test, result, resolution.

Table 5. Concern, test, result and resolution

Concern	Test in paper or review	Result	Resolved?
Is the rescaled intensity C^T tight, so the intensity route works?	Stated in the discussion; footnote 4	Not tight; intensities need not converge in the Skorohod topology	No: the route is abandoned, so the theorem says nothing about intensities
Boundedness of ϕ	Assumptions are weaker than in the 2015 paper	ϕ may blow up at zero, as in order-book fits	Yes, for the theorem
Uniqueness of the solution of (12)	Reference to Mytnik and Salisbury (2015)	Weak uniqueness shown for a similar equation, not this one	No: left open
Limit points versus limit	Tightness gives only subsequences	Statements are for every limit point	Partly: convergence of the whole sequence needs uniqueness
Choice of regime	Heuristic on second moment of C_t^∞	Other regimes give growing variance	Partly: argued, not proved
Finite- T validity	Reviewer's Monte Carlo (Figure 3)	Maximum gap 0.031 in the c.d.f. at $T = 10^4$	Partly: consistent with convergence, rate untested
Randomness of the limit	Reviewer's simulation (Figure 4)	Five-fold spread in total events across seeds	Yes, qualitatively

Source: Jaisson and Rosenbaum (2016), Section 3.4 and footnote 4, and the reviewer's numerical checks.

Two entries deserve emphasis. The tightness failure means that the financial reading of the theorem, with the Hawkes intensity as spot volatility, is not what is proved: what is proved concerns $Y = dX/dt$, the derivative of the limit of the *counts*. The uniqueness gap means that, for the whole sequence, convergence to a single law is not established. Neither invalidates the stated theorems, but both limit the strength of the microstructure interpretation.

7 Critical Evaluation

7.1 What the paper actually proves

ESTABLISHED

For the stated triangular array, the rescaled count and martingale are tight (Proposition 3.2); every limit point satisfies the Volterra equation (11) with the Mittag-Leffler kernel and $Z = B_X$; X has regularity $(1 \wedge 2\alpha) - \varepsilon$; and for $\alpha > 1/2$ the derivative Y solves (12) and has regularity $\alpha - 1/2 - \varepsilon$. The proofs reconstructed above contain no gap that the reviewer could identify.

SUPPORTED BUT CONDITIONAL

That the limit is a fractional CIR process with Hurst parameter $\alpha - 1/2$. The equation is a Volterra equation with a kernel singular like $x^{\alpha-1}$; the fractional Brownian interpretation is an analogy with the representation of Mandelbrot and Van Ness (1968), and it holds up to the uniqueness of the solution. The result also depends on the coupled regime of Assumption 2.

SUGGESTIVE

That order-splitting and high endogeneity cause the roughness of market volatility. The mechanism is consistent with the persistence of signed order flow and the near-one kernel norms cited, but no data are used, and rough volatility could arise in other ways.

NOT ESTABLISHED

That the intensity of the Hawkes process converges to the volatility process; that real markets satisfy Assumption 2; that α and H are linked by $H = \alpha - 1/2$ in finite samples; that the roughness is causal in the sense of a structural decomposition; any statement for $\alpha \leq 1/2$ beyond the regularity of X .

7.2 Strengths

RECOGNISED BY THE AUTHORS

The result is novel in the mathematics of Hawkes processes, since the limit is not a semimartingale. The assumptions are weaker than in the 2015 paper because ϕ need not be bounded. The result offers an agent-based account of a stylised fact.

IDENTIFIED INDEPENDENTLY

- The theorem isolates a single parameter, α , that sets the roughness, and it makes a sharp prediction, $H = \alpha - 1/2$, which can be tested by estimating the two sides independently. This gives the result empirical content despite being a probability paper.
- The threshold $\alpha = 1/2$ is not an artefact of technique: it is where $f^{\alpha,\lambda} \in L^2_{\text{loc}}$ fails, the condition under which the stochastic Volterra integral $\int f(t-s)\sqrt{Y_s}dB_s$ makes sense. The result is internally coherent.
- The proof architecture (tightness, an identity for subsequential limits, a Hölder bootstrap, fractional calculus) is reusable, and it is closely related to the approach of later rough Heston work (El Euch and Rosenbaum 2019).
- The Mittag-Leffler kernel arising from a geometric sum of Pareto lags is a clean, checkable mathematical object. The numerical check in this review reproduces its qualitative behaviour with a few lines of code.

7.3 Weaknesses and critique

CRITIQUE Each item follows the pattern problem, why it matters, potential bias, test.

SINGLE MOST IMPORTANT WEAKNESS: THE COUPLED ASYMPTOTIC REGIME

Problem. Assumption 2 ties a_T and μ_T to the horizon T : $1 - a_T \sim \lambda \delta T^{-\alpha}$ and $\mu_T \sim \mu^* \delta^{-1} T^{\alpha-1}$. A real market has one kernel and one observation window; there is no sequence. **Why it matters.** The theorem is a statement about a family of ever-more-critical processes, and the roughness appears only if the system is tuned to criticality at the rate $T^{-\alpha}$. In a fixed market, whether the observation window lies in the asymptotic regime depends on the unobserved λ . **Potential bias.** If the true kernel norm is, say, 0.9 rather than 0.99, the window in Table 2 is far shorter and the market behaves classically at longer horizons; claiming roughness at all horizons would be an over-reach. **How to test.** Simulate the fixed-parameter process at many horizons and estimate the Hölder exponent of the integrated count's derivative as a function of window length; compare with the empirical scaling of realised variance increments.

OTHER WEAKNESSES

- **Intensity is not volatility.** Problem: the theorem concerns dX/dt , the density of event counts, while volatility concerns price changes. Why it matters: the proportionality of order flow and integrated variance is taken from Wyart et al. (2008), an empirical relation of limited generality across assets and regimes. Potential bias: if variance depends on signed and sized flow, not counts, the roughness of counts may differ from that of variance. Test: regress realised variance on counts and on signed volume over windows.
- **Counts ignore sign and size.** Problem: the model is univariate; the persistence of the *sign* of order flow, which is the empirical long memory, is not represented. The paper links the heavy tail to order splitting, yet a univariate Hawkes process does not distinguish buy from sell. Why it matters: a bivariate or marked model is needed to connect the kernel tail to metaorder lengths. Bias: the mapping from the persistence of signs to α may alter the exponent. Test: derive the limit for a symmetric two-dimensional Hawkes process.
- **Nonstationary limit.** Problem: the process starts from an empty past, so $\mathbb{E}[Y_t] = I^{\alpha, \lambda}(t)$ is increasing from zero. Why it matters: real volatility is stationary with mean reversion. Bias: roughness at the origin of a ramp may not represent roughness in stationary regimes. Test: the stationary version for $\|\phi\|_1 < 1$ with a two-scale analysis.
- **Hölder regularity versus the Hurst exponent measured.** Problem: the empirical H in Gatheral et al. (2018) is estimated from the scaling of moments of log-volatility increments, a different object from the pathwise Hölder exponent of Y , whose derivation is up to an unspecified ε . Why it matters: the equality $H = \alpha - 1/2$ is a statement about exponents of two different functionals. Bias: moment scaling may differ from Hölder regularity under multifractality or with a boundary at zero ($Y_0 = 0$, $Y \geq 0$). Test: compute the structure-function scaling of simulated Y and compare.
- **No uniqueness and no rate.** Problem: weak uniqueness of (12) is open, and no rate of convergence is given. Why it matters: the finite-sample relevance of the limit is unquantified. Bias: Figure 3 shows a visible finite- T gap. Test: derive or simulate the rate.
- **Tail index estimation.** Problem: the theorem assumes an exact regularly varying tail of $1 - F$, whereas empirical kernels are estimated non-parametrically, with truncation at the largest lag used. Why it matters: the tail behaviour beyond the estimation window is not identified, so α is poorly identified from data. Bias: the estimated α may be biased towards values that imply

smoother or rougher volatility. Test: parametric truncated-Pareto fits versus non-parametric ones, with confidence intervals.

7.4 Statistical versus economic significance

CRITIQUE The distinction does not apply to the theorem, which has no estimation error. It applies to the application: a reader should ask whether $H = 0.1$ and $H = 0.25$ imply materially different option prices or hedges. In models of the rough Heston class the short-maturity at-the-money skew scales like $T^{H-1/2}$, so the difference between $H = 0.1$ and $H = 0.25$ is $T^{-0.4}$ against $T^{-0.25}$, a large difference at short maturities. This is a stylised statement used to indicate relevance (the reviewer's inference from the rough volatility literature, not a result of this paper).

8 Contradictions and Edge Cases

CRITIQUE The reviewer tries to break the result by direct reasoning and short calculations.

WHEN DOES THE RESULT DISAPPEAR?

Light tail. If $\int s\phi(s)ds < \infty$, Assumption 1 fails and the earlier theorem gives a Brownian CIR limit with $H = 1/2$ for the derivative; the roughness is therefore not a generic feature of near-instability but of the tail. **Tail index at or below one half.** For $\alpha \leq 1/2$, $f^{\alpha,\lambda} \notin L^2_{\text{loc}}$ near the origin ($\int_0 x^{2(\alpha-1)}dx$ diverges iff $\alpha \leq 1/2$), the stochastic integral in (12) is undefined and X has regularity only $2\alpha < 1$: the limit is not a differentiable activity process. As $\alpha \downarrow 1/2$ the exponent $H \downarrow 0$ and the process degenerates in regularity.

OTHER MECHANISMS GENERATING THE SAME EXPONENT

INFERENCE Roughness near 0.1 can be generated without a Hawkes process: by a fractional Brownian motion with $H \approx 0.1$ driving log-volatility (as in Gatheral et al. 2018), by a volatility process with a kernel singular like $x^{H-1/2}$ in a stochastic Volterra model (Abi Jaber, Larsson and Pulido 2019), or by market-impact kernels with power-law decay (Bouchaud, Gefen, Potters and Wyart 2004, in the library of this project). The theorem shows that one microscopic mechanism is sufficient, not that it is the operative one. Discriminating requires a prediction that differs across mechanisms, for example the joint dynamics of event counts and variance.

EXTREME VALUES AND PARAMETER REGIONS

Large λ . As $\lambda \rightarrow \infty$ the kernel $f^{\alpha,\lambda}$ concentrates near zero (the family tree dies out within a short time) and the limit approaches a law with fast mean reversion; the variance coefficient $1/\sqrt{\mu^*\lambda}$ in (12) vanishes, so the limit becomes deterministic, which is the stable, low-noise regime. **Small λ .** As $\lambda \rightarrow 0$ the process is critical: $F^{\alpha,\lambda}(t) \approx \lambda t^\alpha/\Gamma(1+\alpha) \rightarrow 0$ and the noise coefficient diverges as $\lambda^{-1/2}$, consistent with the discussion of the $v_\infty = 0$ regime, which the authors exclude because the second

moment grows. **Positivity.** $Y \geq 0$ with $Y_0 = F^{\alpha, \lambda}(0) = 0$, so the process begins on the boundary where the square root has infinite slope; this is the same difficulty that makes uniqueness hard.

AN ILLUSTRATIVE CALCULATION: THE BOOTSTRAP

CRITIQUE The bootstrap $H_{n+1} = \alpha + H_n/2$ converges geometrically: for $\alpha = 0.6$, $H_0 = 0.6$, $H_1 = 0.9$, $H_2 = 1$ (capped), so only two iterations are needed for X to reach regularity one; for $\alpha = 0.4$, $H_0 = 0.4$, $H_1 = 0.6$, $H_2 = 0.7$, $H_3 = 0.75$, tending to $0.8 = 2\alpha$. The calculation confirms that the exponent 2α is the unique fixed point below one, so the result is not sensitive to the starting value, but it also shows that the bootstrap carries no information about the roughness of Y , which is obtained only at the last step from the singularity of $D^\nu f^{\alpha, \lambda}$.

9 Further Literature

INFERENCE The connections below follow the pattern paper, shared idea, relationship, importance. Only relations that bear on the content of the theorem are listed. Works that the paper itself cites are marked by their appearance in its bibliography; the later literature is stated from the reviewer's knowledge and from the sources named in the reference list.

Table 6. Literature connections

Paper	Shared idea	Relationship	Why it matters
Hawkes (1971); Hawkes and Oakes (1974)	Self-exciting point process; cluster (branching) representation	Foundation	Defines the object and gives the exact simulation scheme used in the review
Brémaud and Massoulié (2001)	Hawkes processes with $\ \phi\ _1 = 1$ and heavy-tailed kernels	Foundation for the critical case	Shows stationary critical Hawkes processes need $\mu = 0$ and a heavy tail
Bacry, Delattre, Hoffmann and Muzy (2013)	Limit theorems for stable Hawkes; resolvent identities	Technical tool	Supplies the first-moment bounds and the intensity identity used in the proofs
Jaisson and Rosenbaum (2015)	Nearly unstable limit with a light tail	Direct predecessor; contrast case	Brownian CIR limit; the present paper is its heavy-tailed counterpart
Filimonov and Sornette (2012); Hardiman, Bercot and Bouchaud (2013)	Order-flow Hawkes fits with norm near one	Empirical motivation	Source of the near-criticality assumption; calibration issues are documented
Lillo and Farmer (2004); Bouchaud, Gefen, Potters and Wyart (2004)	Long memory of signed order flow and propagator price impact	Microstructure motivation for the heavy tail	Gives a mechanism for $\alpha < 1$, but in the sign process
Gatheral, Jaisson and Rosenbaum (2018)	Realised volatility is rough, $H \approx 0.1$	Empirical target of the theorem	Defines the stylised fact that the paper aims to explain
Mytnik and Salisbury (2015)	Uniqueness for Volterra-type SDEs	Technical neighbour; partial	Closest result to the open uniqueness question

Paper	Shared idea	Relationship	Why it matters
El Euch and Rosenbaum (2019)	Rough Heston model; characteristic function via a fractional Riccati equation	Extension to pricing	The Mittag-Leffler kernel reappears as the resolvent; link to derivatives
Abi Jaber, Larsson and Pulido (2019)	Affine Volterra processes with singular kernels	Generalisation of the limit equation	Existence, uniqueness and affine transforms for Volterra square-root processes
Bayer, Friz and Gatheral (2016)	Pricing under rough volatility	Application	Shows the practical consequences of $H \approx 0.1$

Source: reviewer's synthesis; relations marked as the paper's own references come from its bibliography, and later works are cited from their publication records.

10 Research Gaps

10.1 Paper-level gaps

Gap. The intensity does not converge, so the paper does not characterise the process that actually drives events. **Why it matters.** Estimation from data uses the intensity. **Existing limitation.** Tightness fails. **Way forward.** Study the convergence of a smoothed or integrated intensity, or a different topology, and identify what object corresponds to spot variance.

Gap. Uniqueness of equation (12) and convergence of the whole sequence. **Why it matters.** Without it the statement is about limit points. **Existing limitation.** The singular kernel and the square root block standard Lipschitz and Yamada–Watanabe arguments. **Way forward.** Methods for affine Volterra equations (Abi Jaber, Larsson and Pulido 2019).

Gap. No rate of convergence. **Why it matters.** Figure 3 shows a finite- T gap of three percentage points at $T = 10^4$. **Existing limitation.** The Tauberian remainder is $o(z^\alpha)$ with no explicit rate. **Way forward.** Second-order Tauberian expansions under stronger assumptions on $1 - F$, or direct bounds on Wasserstein distances.

10.2 Literature-level gaps

Gap. Whether order-flow persistence causes rough variance. **Why it matters.** The microstructure explanation competes with other generators of roughness. **Existing limitation.** The theorem is a sufficiency result, and empirical tests of $H = \alpha - 1/2$ across assets are scarce. **Way forward.** A joint cross-sectional estimation of both exponents (Section 11).

Gap. Stationary heavy-tailed near-critical regimes with mean reversion. **Why it matters.** Observed variance is stationary. **Existing limitation.** The limit here starts from an empty past. **Way forward.** Add a baseline that restores stationarity, as in the rough Heston construction.

10.3 Methodological gaps

Gap. Identification of α and $\|\phi\|_1$ from finite samples. **Why it matters.** Both enter the scaling window. **Existing limitation.** Non-parametric kernel estimation with truncation (Bacry, Jaisson and Muzy 2014) is sensitive to the largest lag. **Way forward.** Likelihood-based estimation with a parametric Pareto or Mittag-Leffler kernel, with simulated confidence regions from the cluster representation.

11 Research Opportunities

EXTENSION The following ideas follow from the gaps above. Data sources named are those usually used in this literature; availability and licensing are not verified in this review.

Idea 1. A cross-sectional test of the identity $H = \alpha - 1/2$

RESEARCH QUESTION. Across stocks and years, does the Hurst exponent of realised variance equal the kernel tail index of the order-flow Hawkes process minus one half?

MOTIVATION. The identity is the only direct empirical prediction of the theorem and is stated but not tested.

HYPOTHESIS. $\hat{H}_i = \hat{\alpha}_i - 1/2$ in the cross-section, with slope one and zero intercept in a regression of $\hat{H}_i$ on $\hat{\alpha}_i$.

ECONOMIC MECHANISM. Order splitting creates a power-law kernel, and near-critical endogeneity aggregates it into rough activity.

LITERATURE CONNECTION. Jaisson and Rosenbaum (2016); Gatheral et al. (2018); Bacry, Jaisson and Muzy (2014).

DATA. Tick-level trades and quotes for several hundred stocks, for example from an exchange feed or a commercial database; roughly one decade.

VARIABLES. Kernel tail index $\hat{\alpha}$, kernel norm $\|\hat{\phi}\|_1$, $\hat{H}$ from the scaling of log-variance increments, firm characteristics.

METHOD. Non-parametric and parametric Hawkes estimation; moment-scaling estimation of H on realised variance; errors-in-variables regression with bootstrap standard errors.

IDENTIFICATION STRATEGY. Exogenous variation in order-splitting across stocks (tick size changes, liquidity) shifts α ; test whether H moves one-for-one.

EXPECTED CONTRIBUTION. First direct test of the microstructure foundation of rough volatility.

MAIN RISK. Both estimates are noisy and biased by microstructure noise, so the slope could be attenuated toward zero.

Idea 2. When does roughness appear? A finite-horizon simulation study

RESEARCH QUESTION. For fixed Pareto kernels with realistic $(\alpha, \|\phi\|_1)$, at which horizons does the integrated count behave like the limit of Theorem 3.2?

MOTIVATION. Table 2 suggests windows of hundreds to thousands of kernel units; whether intraday horizons fall inside is unknown.

HYPOTHESIS. The estimated Hölder exponent of Y converges to $\alpha - 1/2$ only when the window exceeds $(1 - \|\phi\|_1)^{-1/\alpha}$ and is close to $1/2$ below it.

ECONOMIC MECHANISM. Same as the paper; the question is the crossover scale.

LITERATURE CONNECTION. Jaisson and Rosenbaum (2015, 2016); Hardiman et al. (2013).

DATA. Simulated; calibrated to published kernel estimates.

VARIABLES. α , $\|\phi\|_1$, T , estimated roughness index.

METHOD. Cluster simulation, grid over parameters, structure-function estimator of H with bootstrap bands.

IDENTIFICATION STRATEGY. Not applicable (controlled experiment); parameters are known by construction.

EXPECTED CONTRIBUTION. A quantitative map of the asymptotic regime and a rate of convergence conjecture.

MAIN RISK. Computational cost near criticality, where the number of events grows like $T^{2\alpha}$.

Idea 3. A sign-aware limit for buy and sell order flow

RESEARCH QUESTION. What is the scaling limit of a symmetric two-dimensional nearly unstable Hawkes process with heavy-tailed self and cross kernels, and does it reproduce rough variance with a price process of the right drift?

MOTIVATION. The empirical long memory is in the sign of orders, not the counts.

HYPOTHESIS. Variance inherits $H = \alpha - 1/2$ and the signed flow inherits a long-memory exponent consistent with metaorder lengths.

ECONOMIC MECHANISM. Metaorders split into child orders of the same sign; sign persistence and activity clustering arise jointly.

LITERATURE CONNECTION. Lillo and Farmer (2004); Tóth et al. (2011); Bacry et al. (2013).

DATA. Signed trades with account or metaorder identifiers where available.

VARIABLES. Sign autocorrelation exponent, kernel tails, price impact exponent.

METHOD. Matrix-valued resolvent and Laplace analysis; Skorokhod tightness for the vector process.

IDENTIFICATION STRATEGY. Match the three exponents (sign memory, activity roughness, impact decay) through one parameter.

EXPECTED CONTRIBUTION. A unified theory of long memory in signs and roughness in volatility.

MAIN RISK. Technically demanding; tightness of the vector process may fail.

Idea 4. Does market-implied roughness match order-flow roughness?

RESEARCH QUESTION. Do the roughness parameters implied by short-dated at-the-money skews equal those estimated from order flow, for the same underlying and period?

MOTIVATION. Options embed risk-neutral expectations; if the Hawkes mechanism is real, both worlds should share H .

HYPOTHESIS. The skew scaling exponent $H_Q - 1/2$ is consistent with $\hat{\alpha} - 1$ from order flow.

ECONOMIC MECHANISM. The physical roughness is transmitted to implied volatility through the pricing kernel; a risk premium may shift levels but not exponents.

LITERATURE CONNECTION. El Euch and Rosenbaum (2019); Bayer, Friz and Gatheral (2016).

DATA. Index and single-stock option surfaces plus matched tick data.

VARIABLES. Skew term structure exponent, $\hat{\alpha}$, $\hat{H}$, maturities from one day.

METHOD. Power-law fits of the skew with confidence bands; comparison of exponents by Wald tests.

IDENTIFICATION STRATEGY. Event-day variation (announcements) shifting order-flow persistence.

EXPECTED CONTRIBUTION. Bridges microstructure and derivatives pricing.

MAIN RISK. Skew measurement at ultra-short maturities is noisy.

Idea 5. Time variation of criticality and roughness

RESEARCH QUESTION. Do $\|\phi\|_1$ and α vary across market regimes, and does H_t move accordingly?

MOTIVATION. The theorem requires near criticality; endogeneity is likely to increase in stress periods.

HYPOTHESIS. Stress raises $\|\phi\|_1$ and lengthens the roughness window; $\hat{H}_t$ declines when $\hat{\alpha}_t$ declines.

ECONOMIC MECHANISM. Feedback between liquidity provision and demand intensifies in stress.

LITERATURE CONNECTION. Hardiman et al. (2013); Filimonov and Sornette (2012).

DATA. Tick data across crisis and calm periods, for example 2008, 2020 and 2022.

VARIABLES. Rolling $\|\hat{\phi}\|_1$, $\hat{\alpha}$, $\hat{H}$, VIX.

METHOD. Rolling Hawkes estimation with change-point tests.

IDENTIFICATION STRATEGY. Compare crisis and non-crisis windows within stock.

EXPECTED CONTRIBUTION. Links criticality to measurable regime variation.

MAIN RISK. Short windows give noisy estimates; non-stationarity within windows.

Table 7. Ranking of research ideas (scores from 1 to 5; reviewer's judgement)

Idea	Novelty	Feasibility	Data	Method	Importance	Total
1. Cross-sectional test of $H = \alpha - 1/2$	4	4	4	4	5	21
2. Finite-horizon simulation study	3	5	5	4	3	20
4. Implied versus order-flow roughness	4	3	3	4	4	18
3. Sign-aware limit theorem	5	2	3	3	4	17
5. Time-varying criticality	3	3	4	3	3	16

Source: reviewer's scoring; subjective, with no external validation.

EXTENSION The top three are ideas 1, 2 and 4. Idea 1 tests the single prediction that gives the theorem empirical content and can be done with standard estimators. Idea 2 is cheap, needs no data licences and is a prerequisite for interpreting any estimate from idea 1, since it tells which window and which estimator to trust. Idea 4 extends the test to the derivatives market, where the economic stakes are highest. Idea 3 is the most novel but least likely to succeed within a single project.

12 Learning Framework

12.1 Prerequisite concepts

- Point processes, compensators and the martingale $N - \int \lambda$; branching processes with immigration.
- Convolution, Laplace transform, and the Tauberian theorem of Karamata for regularly varying functions (Bingham, Goldie and Teugels 1989).
- Weak convergence on the Skorokhod space, tightness, and the criteria of Jacod and Shiryaev (2003).
- Fractional integrals and derivatives, Hölder regularity, and the Mittag-Leffler function (Samko, Kilbas and Marichev 1993).
- The CIR process, and stochastic Volterra equations; fractional Brownian motion in its Mandelbrot and Van Ness representation.

12.2 Core concepts learned

- A norm near one together with a heavy tail produces a non-semimartingale limit.
- Aggregation of persistent shocks smooths the count but roughens its derivative.
- The Mittag-Leffler function is the resolvent of a Hawkes process with a regularly varying kernel in the near-critical scaling.
- The threshold $\alpha = 1/2$ is the square-integrability limit of the kernel.

12.3 Important equations

Equations (1), (5) and (8) for the mechanism; Assumption 2 for the regime; (11) and (12) for the results; and the bootstrap map $H \mapsto \alpha + H/2$ for the proof.

12.4 Methods learned

Tightness plus identification of limit points; Skorokhod representation; the Dambis–Dubins–Schwarz time change; fractional calculus to differentiate a Volterra convolution; stochastic Fubini; Hölder bootstrapping; cluster simulation of Hawkes processes.

12.5 Questions to test understanding

COMPREHENSION

1. What does $\|\phi\|_1$ measure economically, and why does a value near one correspond to a market in which most orders are endogenous?
2. Why is the law of large numbers for N^T lost when $a_T \rightarrow 1$?
3. What is $f^{\alpha,\lambda}$, and where does its singularity at zero come from?

ANALYSIS

1. Derive $\hat{\rho}^T(z) \sim v_T/(v_T + z^\alpha)$ from the Laplace transform of a geometric sum and the Karamata expansion.
2. Why must ν lie in $(1/2, \alpha)$ in the proof of Theorem 3.2, and what happens to the argument at $\alpha \leq 1/2$?
3. Why does the bootstrap $H \mapsto \alpha + H/2$ stop at $2\alpha \wedge 1$, and why is it not enough for the roughness of Y ?

RESEARCH

1. Which observable quantity could discriminate the Hawkes explanation of roughness from a fractional-noise explanation?
2. What changes in the limit if the kernel is a mixture of a light-tailed and a heavy-tailed component?
3. Is the intensity of a heavy-tailed nearly unstable Hawkes process convergent under a topology weaker than Skorokhod's J_1 ?

12.6 Papers to read next

El Euch and Rosenbaum (2019) for the pricing consequences; Abi Jaber, Larsson and Pulido (2019) for the general Volterra framework; Bacry, Jaisson and Muzy (2014) for the estimation of slowly decaying kernels; and Bayer, Friz and Gatheral (2016) for rough-volatility pricing.

13 Conclusion

13.1 One-sentence thesis

Heavy-tailed, nearly unstable Hawkes processes converge to a random, non-semimartingale limit whose rate of activity is a fractional CIR process of Hurst parameter $\alpha - 1/2$, so persistent order flow together with high endogeneity can generate rough volatility.

13.2 Three most important insights

- Fat tails mean persistence in the kernel but roughness in the limit: aggregation, not persistence, sets the exponent.
- The roughness exponent is tied to a single microscopic parameter, $H = \alpha - 1/2$, and exists only for $\alpha > 1/2$.
- The result holds in a coupled regime near criticality; its relevance for a given market depends on whether the observation window exceeds $(1 - \|\phi\|_1)^{-1/\alpha}$.

13.3 Strongest evidence

The proofs of Theorems 3.1 and 3.2, together with the reviewer's check that the law of the family time converges towards the Mittag-Leffler distribution and that simulated sample paths remain random at first order.

13.4 Biggest weakness

The coupled asymptotic regime: the theorem needs the kernel norm and the baseline intensity to scale with the horizon, which a single real market does not provide, and the paper does not show that empirical windows lie in the asymptotic range.

13.5 Most important research gap

An empirical test of the identity $H = \alpha - 1/2$ with independent estimates of both sides.

13.6 Best research opportunity

The cross-sectional test of Idea 1, supported by the finite-horizon simulation map of Idea 2.

13.7 Three papers to read next

El Euch and Rosenbaum (2019); Abi Jaber, Larsson and Pulido (2019); Bacry, Jaisson and Muzy (2014).

14 References

- Abi Jaber, E., Larsson, M., and Pulido, S. (2019). Affine Volterra processes. *The Annals of Applied Probability*, 29(5), 3155–3200.
- Bacry, E., Delattre, S., Hoffmann, M., and Muzy, J.-F. (2013). Some limit theorems for Hawkes processes and application to financial statistics. *Stochastic Processes and their Applications*, 123(7), 2475–2499.

- Bacry, E., Jaisson, T., and Muzy, J.-F. (2014). Estimation of slowly decreasing Hawkes kernels: application to high frequency order book modelling. Working paper.
- Bayer, C., Friz, P., and Gatheral, J. (2016). Pricing under rough volatility. *Quantitative Finance*, 16(6), 887–904.
- Bingham, N. H., Goldie, C. M., and Teugels, J. L. (1989). *Regular Variation*. Cambridge University Press.
- Bouchaud, J.-P., Gefen, Y., Potters, M., and Wyart, M. (2004). Fluctuations and response in financial markets: the subtle nature of random price changes. *Quantitative Finance*, 4(2), 176–190.
- Brémaud, P., and Massoulié, L. (2001). Hawkes branching point processes without ancestors. *Journal of Applied Probability*, 38(1), 122–135.
- El Euch, O., and Rosenbaum, M. (2019). The characteristic function of rough Heston models. *Mathematical Finance*, 29(1), 3–38.
- Filimonov, V., and Sornette, D. (2012). Quantifying reflexivity in financial markets: toward a prediction of flash crashes. *Physical Review E*, 85(5), 056108.
- Gatheral, J., Jaisson, T., and Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- Hardiman, S. J., Bercot, N., and Bouchaud, J.-P. (2013). Critical reflexivity in financial markets: a Hawkes process analysis. *The European Physical Journal B*, 86(10), 1–9.
- Hawkes, A. G. (1971). Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1), 83–90.
- Hawkes, A. G., and Oakes, D. (1974). A cluster process representation of a self-exciting process. *Journal of Applied Probability*, 11(3), 493–503.
- Jacod, J., and Shiryaev, A. N. (2003). *Limit Theorems for Stochastic Processes*, 2nd ed. Springer.
- Jaisson, T., and Rosenbaum, M. (2015). Limit theorems for nearly unstable Hawkes processes. *The Annals of Applied Probability*, 25(2), 600–631.
- Jaisson, T., and Rosenbaum, M. (2016). Rough fractional diffusions as scaling limits of nearly unstable heavy tailed Hawkes processes. *The Annals of Applied Probability*, 26(5), 2860–2882. DOI 10.1214/15-AAP1164.
- Lillo, F., and Farmer, J. D. (2004). The long memory of the efficient market. *Studies in Nonlinear Dynamics and Econometrics*, 8(3), Article 1.
- Mandelbrot, B. B., and Van Ness, J. W. (1968). Fractional Brownian motions, fractional noises and applications. *SIAM Review*, 10(4), 422–437.
- Mytnik, L., and Salisbury, T. S. (2015). Uniqueness for Volterra-type stochastic integral equations. arXiv preprint 1502.05513.
- Revuz, D., and Yor, M. (1999). *Continuous Martingales and Brownian Motion*, 3rd ed. Springer.
- Samko, S. G., Kilbas, A. A., and Marichev, O. I. (1993). *Fractional Integrals and Derivatives: Theory and Applications*. Gordon and Breach.
- Tóth, B., Lemperiere, Y., Deremble, C., de Lataillade, J., Kockelkoren, J., and Bouchaud, J.-P. (2011). Anomalous price impact and the critical nature of liquidity in financial markets. *Physical Review X*, 1(2), 021006.
- Veraar, M. (2012). The stochastic Fubini theorem revisited. *Stochastics*, 84(4), 543–551.

Wyart, M., Bouchaud, J.-P., Kockelkoren, J., Potters, M., and Vettorazzo, M. (2008). Relation between bid–ask spread, impact and volatility in order-driven markets. *Quantitative Finance*, 8(1), 41–57.

Zhu, L. (2013). Nonlinear Hawkes processes. arXiv preprint 1304.7531.